P R O J E C T   P R O P O S A L

Dengue Outbreak Risk Prediction

Predicting subzone-level dengue cluster risk 14 days ahead using Singapore government open data

Module	CS611 — Machine Learning Engineering
Institution	Singapore Management University, SCIS
Submission	Week 6 Project Proposal
Domain	Public Health · Geospatial ML · MLOps

1. Executive Summary

Dengue fever remains one of Singapore's most persistent public health challenges. In 2020, Singapore recorded over 35,000 confirmed cases — the largest outbreak in the country's recorded history — driven by a serotype shift from DENV-1/2 to DENV-3, which collapsed population immunity that had built up over the preceding decade. The National Environment Agency (NEA) currently deploys fogging and inspection teams reactively — after clusters are declared — leaving a critical 7–14 day window during which Aedes aegypti mosquitoes are actively breeding and transmitting the virus uninterrupted.

This project proposes an end-to-end machine learning pipeline that predicts which of Singapore's 330 residential subzones will have active dengue clusters in the following 14 days, enabling NEA and licensed pest control operators to pre-deploy fogging teams before outbreaks form. The pipeline ingests multiple Singapore government open datasets, engineers epidemiologically-grounded features, trains a binary classifier using time-series-safe cross-validation, and serves weekly batch predictions through an Airflow-orchestrated production system running entirely in Docker.

2. Business Problem

2.1 Problem Statement

The Aedes aegypti mosquito's breeding cycle takes 7–14 days from egg to adult. This means that by the time NEA declares a dengue cluster — which requires confirmed human cases — the mosquito population has already been established for at least one full breeding cycle. Reactive fogging at this stage reduces, but cannot eliminate, the outbreak. The operational opportunity is therefore at the pre-cluster stage: if areas of elevated risk can be identified two weeks in advance, fogging and larval source removal can be deployed while the mosquito population is still in early development.

This is a forward-looking prediction problem. The question is not 'where are the clusters now?' — that is already answered by NEA's public cluster map — but rather 'where will clusters form next?' based on weather conditions, population density, historical cluster patterns, and seasonal signals.

2.2 Business Impact

The direct beneficiaries of accurate 14-day outbreak forecasts are twofold. First, NEA's Vector Control Operations Division, which plans weekly fogging schedules and can pre-allocate resources to predicted high-risk subzones. Second, commercial pest control operators such as Rentokil and PestBusters, who contract with NEA and private estates and benefit from demand-anticipation to pre-position teams and consumables.

Beyond operational efficiency, there is a public health argument: the World Health Organisation estimates dengue costs Southeast Asia approximately USD 8.6 billion annually in treatment costs and lost productivity. A 14-day predictive window, even with moderate accuracy, materially shifts the economics of outbreak control from expensive reactive treatment to cheaper preventive intervention.

2.3 Success Criteria

The model is not evaluated on prediction accuracy alone. Consistent with the course's emphasis on production ML engineering, success is defined across three dimensions:

- Prediction quality: Recall ≥ 0.70 for High-risk subzones on held-out 2020 data (optimising recall over precision because false negatives — missed outbreaks — carry higher public health cost than false positives — over-deployed fogging teams).
- Pipeline robustness: End-to-end pipeline runs reliably in Docker on a local machine with a single command, with all intermediate data versioned and traceable.
- Monitoring completeness: Automated alerts fire when model performance degrades below threshold, with a documented retraining trigger protocol.

3. Dataset

3.1 Data Architecture — Medallion Model

The pipeline organises data into three layers following the Medallion Architecture taught in this course. This separation ensures reproducibility, auditability, and clean handoffs between pipeline stages.

Layer	Purpose	Contents for this project
🥉 **Bronze**	*Raw Integration — Landing zone, no schema enforcement*	Raw cluster snapshots (SGCharts CSV), raw weather readings (NEA API), raw MRT tap volumes (LTA), raw population counts (SingStat). Ingested as-is, no transformation.
🥈 **Silver**	*Filtered, Cleaned, Augmented — Schema enforced, quality assured*	Validated and deduplicated cluster records, nulls handled, weather readings aggregated to daily mean per station, population joined to URA subzone grid, datasets joined on date and subzone.
🥇 **Gold**	*Feature Store — Business-level aggregates, ML-ready*	Engineered features per subzone per week: rainfall lag 1/2/4 weeks, temperature rolling mean, cluster proximity score, population density weight, seasonality features. Binary label: cluster present in next 14 days (1/0).

3.2 Data Sources

Source	Role	Coverage	Key Fields
SGCharts Dengue Clusters	Anchor	May 2013 – Nov 2020	Cluster lat/lng, case count, snapshot date. 250+ bi-weekly snapshots. Community-maintained CSV from NEA website scraping.
NEA Historical Daily Weather	Auxiliary 1	2013 – 2017 (CSV), 2018 – 2020 (API)	Daily rainfall total (mm), mean/max/min temperature (°C), mean wind speed. Admiralty station (CSV); 30+ stations via NEA v2 API.
NEA v2 Real-time API	Auxiliary 1 ext.	Dec 2016 – present	api-open.data.gov.sg/v2/real-time/api — rainfall, air-temperature, relative-humidity. Paginated JSON, no API key required.
LTA MRT Tap Volumes	Auxiliary 2	2013 – 2020 (monthly)	Monthly tap-in/tap-out by station. Used as neighbourhood activity proxy. Requires free LTA DataMall API key.

Source	Role	Coverage	Key Fields
SingStat Subzone Population	Auxiliary 3	2010 and 2020 Census	Resident population by planning area and subzone. Used to compute exposure risk weight per subzone.
URA Subzone Boundaries	Spatial reference	Master Plan 2019	GeoJSON polygon boundaries for 330 Singapore subzones. Used for spatial join of cluster coordinates to subzone labels.

3.3 Label Engineering

The training label is binary: does subzone S have an active dengue cluster in the 14 days following observation date T? This is derived by spatially joining SGCharts cluster coordinates (lat/lng) to the URA subzone boundary GeoJSON using GeoPandas, then rolling a 14-day forward window per subzone. A subzone is labelled positive (1) if any cluster centroid falls within its boundary during the window.

This is a documented assumption: SGCharts records cluster centroids, not precise boundaries. Boundary-to-centroid assignment introduces minor spatial error at subzone edges. This is acknowledged as a known limitation and is consistent with the level of spatial precision available in public data.

3.4 Assumptions

- Population data: 2010 Census figures are used for 2013–2014 records; 2020 Census for 2015–2020. Intervening years are linearly interpolated. This is standard practice in spatial epidemiology.
- SGCharts data quality: Community-maintained scrapes are assumed accurate within ±1 day of the recorded snapshot date.
- Weather generalisation: The historical CSV provides data for Admiralty station only (2013–2017). Island-wide generalisation is approximated using Admiralty as the reference station, supplemented by multi-station API data from 2018 onwards.
- Post-2020 gap: SGCharts data ends November 2020. The 2020 dataset is retained in full as it contains the record outbreak and the serotype shift event, which is the primary concept drift demonstration in the monitoring component.

4. Users and System Use

4.1 Primary User — NEA Vector Control Operations

Every Monday morning, NEA's Vector Control Operations Division accesses the system's output: a ranked list of Singapore's 330 subzones scored as Low, Medium, or High risk for the coming 14 days. The team uses this to allocate fogging and inspection resources for the week's schedule. The system does not replace human judgement — it provides a risk signal that is one input into the team's operational planning alongside complaint reports and field observations.

The user does not interact with raw model probabilities. The output is a risk tier and, optionally, the top three contributing features per high-risk subzone (e.g. 'rainfall last 2 weeks elevated, prior cluster within 500m, high population density'). This is generated via SHAP values and formatted for non-technical operational staff.

4.2 Secondary User — Pest Control Operators

Licensed pest control companies contracted by NEA and private estates receive the same weekly risk map. They use it to pre-position fogging equipment and personnel in predicted hotspot zones before NEA issues formal fogging requests. This reduces mobilisation time from 24–48 hours to near-zero for predicted clusters.

4.3 Internal User — Data and Operations Team

The internal data team monitors the model via a weekly performance dashboard showing precision, recall, and prediction distribution across subzones. When the Gini coefficient drops below a defined threshold — indicating degraded discriminative ability — an automated alert fires via the Alarm Manager component, and the Scheduler triggers the retraining Airflow DAG. The data team reviews the retrained model before promotion to production.

5. Environment

5.1 Deployment Environment

The pipeline runs entirely on local infrastructure via Docker and docker-compose. No cloud services or paid infrastructure are required. All components — Airflow, Postgres, the training container, and the inference container — are defined in a single docker-compose.yaml and can be launched with one command on any machine with Docker installed. This design decision follows the course guidance that production-grade tooling should be deployable locally at minimum.

The system is architected so that migration to cloud infrastructure (AWS ECS, GCP Cloud Run, or similar) requires only configuration changes — replacing local volume mounts with S3/GCS references and pointing Postgres to a managed database. The application code and DAG definitions are environment-agnostic.

5.2 Technology Stack

Component	Tool	Purpose
Containerisation	Docker + docker-compose	All pipeline components packaged as containers; single-command local deployment
Orchestration	Apache Airflow	Weekly DAG for ingestion, preprocessing, feature engineering, inference, and monitoring
Data Storage	PostgreSQL	Raw storage (Bronze), Silver tables, Gold feature store, and inference output tables
Spatial Processing	GeoPandas + Shapely	Spatial join of cluster coordinates to URA subzone boundaries; subzone grid creation
Model Training	XGBoost / LightGBM	Binary classifier for cluster risk per subzone per week
Hyperparameter Tuning	Optuna	Bayesian hyperparameter optimisation over key model parameters
Experiment Tracking	MLflow	Tracks all training runs, parameters, metrics, and artefacts
Model Registry	MLflow Model Registry	Versions trained models; promotes candidates from Staging to Production
Data Processing	PySpark / Pandas	Large-scale data transformation for Bronze → Silver → Gold transitions
Explainability	SHAP	Per-subzone feature attribution for high-risk predictions
Monitoring	Custom + Airflow	Weekly Gini and Precision/Recall tracking; automated drift alerts

6. High-Level Design

6.1 Pipeline Overview

The pipeline follows the four-phase ML lifecycle structure: Process Data → Develop Model → Deploy → Monitor & Retrain. Each phase is orchestrated by Airflow DAGs and produces well-defined artefacts that feed the subsequent phase.

Data ingestion runs weekly, triggered every Sunday night, so that processed features are available for Monday morning inference. The training pipeline runs on-demand and on drift-triggered retraining events. Batch inference runs every Monday at 06:00 SGT. Monitoring runs immediately after inference completion.

6.2 Feature Engineering Rationale

Feature design is grounded in the epidemiology of dengue transmission. The Aedes aegypti mosquito completes its breeding cycle in 7–14 days under Singapore's tropical conditions, and larvae require standing water — meaning rainfall is the most important environmental predictor. The following features are engineered at the Gold layer:

- Rainfall lag features (1-week, 2-week, 4-week cumulative): captures the lagged effect of rainfall on mosquito population growth. A 2-week lag is the strongest predictor consistent with the breeding cycle.
- Temperature rolling mean (4-week): higher temperatures accelerate the Aedes lifecycle; Singapore's temperature variance is low but meaningful across seasons.
- Relative humidity (7-day mean): influences mosquito survival rate; high humidity extends adult lifespan.
- Cluster proximity score: binary indicator of whether any active cluster existed within a defined radius of the subzone in the prior 4 weeks. Captures spatial autocorrelation of outbreaks.
- Population density weight: resident population per subzone from SingStat, normalised to a 0-1 scale. High-density areas have higher exposure risk for any given mosquito population.
- Seasonality features: week-of-year and month encoded cyclically (sin/cos). Singapore has two dengue peak seasons annually (May–June and October–November).
- Prior cluster history: number of weeks in the past 12 months with an active cluster in the subzone. Captures structural risk factors such as vegetation density and drainage quality.

6.3 Model Selection Rationale

XGBoost (primary) and LightGBM (challenger) are selected over deep learning approaches for three reasons. First, the dataset is tabular with mixed feature types (continuous weather, binary cluster indicators, ordinal temporal features), a domain where gradient boosting consistently outperforms neural approaches without significant hyperparameter investment. Second, interpretability is required: SHAP values on XGBoost/LightGBM produce reliable feature attributions that can be communicated to non-technical NEA operational staff. Third, training time on a local machine is manageable — gradient boosting on a dataset of 330 subzones × 400 weeks fits comfortably in memory and trains in minutes rather than hours.

6.4 Cross-Validation Strategy

Standard k-fold cross-validation is explicitly not used. Because the label (cluster presence in next 14 days) is derived from a forward temporal window, using random k-fold splits would allow future cluster data to leak into training folds — a form of temporal data leakage that produces optimistic and unreliable performance estimates. Instead, time-series cross-validation with a sliding window is used: the model is trained on all data up to week T and evaluated on weeks T+1 through T+4, then the window advances. This mirrors the actual deployment scenario where the model is always trained on past data and evaluated on future data.

6.5 Deployment Strategy

Batch inference is the correct deployment mode for this use case. NEA's fogging schedule is planned weekly, not in real time. There is no operational requirement for sub-minute latency. Batch inference run every Monday produces a week's worth of predictions in a single efficient job, with results stored in Postgres and consumed by the operational dashboard.

A canary deployment strategy is used for model updates. When a new model is promoted to Production in the MLflow registry, it initially scores 20% of subzones in parallel with the incumbent model. If performance metrics on the canary cohort match or exceed the incumbent over two consecutive weeks, the new model is promoted to full production. This minimises the risk of deploying a degraded model to all subzones simultaneously.

6.6 Monitoring and Retraining

The monitoring rationale is grounded in a historically documented concept drift event. In 2020, the predominant dengue serotype in Singapore shifted from DENV-1/2 to DENV-3. This caused population immunity — accumulated through years of DENV-1/2 exposure — to collapse, resulting in the record 35,000-case outbreak. A model trained entirely on pre-2020 data would not have captured the changed dynamics of this serotype, because historical cluster patterns, population susceptibility, and transmission rates all changed simultaneously.

This demonstrates that dengue outbreak prediction is subject to genuine concept drift — not just data distribution shift — and that a monitoring and retraining pipeline is not a theoretical exercise but an operational necessity. The monitoring component tracks three signals weekly: the Gini coefficient of the predicted risk distribution, precision at the High-risk tier, and feature distribution drift (KL divergence on rainfall and temperature inputs). If any signal exceeds its drift threshold, the Alarm Manager fires an alert and the Scheduler queues a retraining job.

7. Open Questions for Feedback

The team invites feedback on the following design decisions before finalising the implementation plan:

- Granularity: the current design scores 330 subzones. An alternative is to aggregate to 55 URA Planning Areas, which reduces sparsity in the label (fewer small subzones with rare cluster events) at the cost of less precise operational guidance. Recommendation sought.
- Post-2020 data gap: SGCharts ceased collection in November 2020. The team proposes to train on 2013–2020 and demonstrate concept drift using 2020 as the drift event. An alternative is to synthesise 2021–2025 cluster data calibrated to NEA's published yearly case totals. Recommendation sought on which approach better demonstrates production readiness.
- Precision vs Recall tradeoff: in public health, false negatives (missed outbreak subzones) carry higher cost than false positives (over-deployed fogging teams). The team proposes to optimise for Recall ≥ 0.70 at the High-risk tier. Is this the right threshold, or should the operational team define it?
- SHAP in production: including per-subzone SHAP explanations in the weekly output adds ~30 seconds to inference time. Is this warranted for the proposal scope, or should it be deferred to a v2 enhancement?

8. References

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